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ORIGINAL ARTICLE

Characterization of the *Anopheles funestus* group, including *Anopheles funestus*-like, from northern Malawi

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Background: Limited information is available on malaria vector composition, feeding habits and malaria transmission in northern Malawi. Evidence of mosquito species diversity in this area was established in 2009, when *Anopheles funestus*-like, a new member of the *An. funestus* group was described. Additional biological information is needed to identify this species and to understand its role in malaria transmission.

Methods: Anopheline mosquitoes were collected in northern Malawi and analyzed for *Plasmodium* species infection, blood meal source and susceptibility to insecticides. A new hydrolysis probe assay was designed to identify *An. funestus*-like.

Results: *Anopheles funestus* and *An. rivulorum* predominated in the indoor collections. Most *An. funestus*-like were collected indoors, mainly fed on animals and were uninfected with *P. falciparum*. *Anopheles funestus* showed insecticide resistance to deltamethrin and bendiocarb. A high-precision hydrolysis probe assay was successfully developed to identify *An. funestus*-like.

Discussion: Four species in the *An. funestus* group were collected in Karonga. Resistance to deltamethrin and bendiocarb was observed in *An. funestus* and further investigation is needed on the insecticide resistance mechanisms. *Anopheles funestus*-like, while collected indoors, is mainly zoophilic and most likely not a malaria vector.

Accession numbers: *An. funestus* (GenBank accession no. **KC771136**), *An. funestus*-like (GenBank accession no. **KC771137**), *An. parensis* GenBank accession no. **KC771138**) and *An. vaneedeni* GenBank accession no. **KC771139**).

Keywords: African malaria vector, Identification, Insecticide resistance

Introduction

In Malawi, malaria is endemic and remains one of the leading causes of mortality and morbidity. The (US) President's Malaria Initiative: Malaria Operational Plan for Malawi[1] estimates that 97% of the country's 13.1 million population is at risk of the disease, with children below the age of 5 years and pregnant women being the most vulnerable.

Malaria transmission in Malawi is perennial but peaks from October to April.[1] Malaria vectors *An. funestus*, *An. gambiae* and *An. arabiensis* are controlled mainly by indoor house spraying (IRS) with residual insecticides or the use of long-lasting insecticide-treated bed nets. Unfortunately, resistance to two classes of insecticides by vector mosquitoes[2,3] presents a challenge to malaria control in Malawi. The situation is further complicated by the recent discovery in Karonga, northern Malawi, of a new mosquito species in the *An. funestus* group, currently referred to as *An. funestus*-like.[4] Knowledge of the behavior and ecology of *An. funestus*-like in terms of its role in malaria transmission is lacking.

The aim of this study was to provide baseline information on malaria vectors and their insecticide susceptibility status in Karonga, northern Malawi, and specifically to establish host preference, *Plasmodium falciparum/vivax* infection rates and resting behavior of *An. funestus*-like. In order to do this an identification assay was developed to differentiate *An. funestus*-like from other morphologically similar species in the group.

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Materials and methods

Description of study sites

We carried out our study in five villages near Karonga, in northern Malawi. This region has a dry season from April to November and a rainy season from December to March. The mean annual rainfall is 1025 mm.[5] Six sites in five villages (Figure 1) were selected for specimen collection on the basis of availability of breeding sites. The villages north of Karonga included: Mwenetete (two sites: lat 9°48.257′S, long 53°52.161′E and lat 9°47.635′S, long 33°52.342′E), Mwakabighili (lat 9°49.65′S, long 33°52.33′E), Kwambwe and Mwampaghatwa (lat 9°54.032′S, long 33°55.792′E). South of Karonga, specimens were collected from Wovwe (lat 10°18.627′S, long 34°07.901′E).

All houses were unsprayed and had on average two bed nets in good condition, most of these being insecticide treated. Following two earthquakes in the region in December 2009, some inhabitants slept in sheds for safety reasons. Domestic animals such as chickens, dogs and cattle are accommodated close to the houses.

Mosquito collections

Adult *Anopheles* mosquitoes resting indoors or outdoors were collected from 21–29 April and from 24 September to 8 October 2010, using hand aspirators. Collections were done in the mornings (08:00–12:00 h) and afternoons (15:00–17:00 h). Houses without nets or those with untreated nets and close to breeding sites were preferred. Outdoor sites included tree holes, empty containers and animal shelters. Human landing catches and cattle kraal collections were performed in Kwambwe village late in the evenings. Live mosquitoes were immobilized using ethyl acetate and individually stored in 0.2 ml Eppendorf tubes containing silica gel (Fisher Scientific, Fair Lawn, NJ, USA). A subset of the female mosquitoes collected was kept alive on 10% sugar solution and transported to Johannesburg, South Africa, for further analysis. The number of collections by different collection methods and respective capture effort (in mosquitoes/collector-hour) varied according to collectors' availability and access to the relevant areas.

Laboratory mosquito rearing and insecticide susceptibility tests

Live wild-caught females were placed in individual oviposition glass vials containing moist filter paper to induce egg laying. Eggs were reared as described by Hunt et al.[6]

WHO susceptibility testing

Reared F_1 adults were maintained on 10% sugar water until used for insecticide susceptibility assays[7] at 2–5 days old. Insecticidal activity of papers was first tested by exposing a susceptible *An. funestus* laboratory strain (FANG).[6] Three replicates of 25 mosquitoes each were exposed to 4% DDT, 0.05% deltamethrin and 0.01% bendiocarb for 1 h each. For each bioassay, control tubes containing 25 mosquitoes were exposed to papers without insecticide. Final mortality was recorded 24 h post exposure. Where the mortality rate of the control was between 5% and 20%, mortality rates for test samples were corrected using Abbot's formula (Abbott, 1925).[8] No control mortality greater than 20% was obtained. Data were analyzed using WHO criteria.[7]



Figure 1. Map of Malawi showing sample collection sites (pins). Obtained from a Google Earth map (http://maps.google.co.za; accessed 13 Aug 2013).

Anopheles funestus group identification

Field-collected mosquitoes were morphologically identified[9,10] as belonging to the *An. funestus* group. DNA was extracted[11] and each mosquito identified by the PCR method of Koekemoer et al.[12] Unidentified samples were then subjected to PCR according to Spillings et al.[4] for identification of *An. funestus*-like.

Anopheles funestus and *An. funestus*-like specimens were molecularly typed using D3 restriction fragment length polymorphism (RFLP) analysis.[13,14] A portion of the D3 PCR product was cloned using pGEM-T Easy Vector kit A1360 (Promega, Madison, WI, USA) following the manufacturer's instructions and sequenced. The sequences were analyzed for *Hpa* II restriction site (CCGG) to explain the observed RFLP patterns. Molecular type identification was according to the naming described by Garros et al. and Koekemoer et al.[13,14]

Determination of *Plasmodium falciparum* and *P. vivax* sprorozoite infection

ELISA (enzyme-linked immunosorbent assay) was used to detect *Plasmodium* species circumsporozoite proteins (CSP) in the head and thorax of wild-caught female mosquitoes.[15] Positive controls provided by the Centers for Disease Control and Prevention (Atlanta, GA, USA) consisted of synthetic peptides for *P. falciparum* and *P. vivax,* and were freshly prepared on the day of the ELISA. Antibodies of the other *Plasmodium* species were not available during this study and were excluded. Negative controls consisted of seven unfed laboratory *An. funestus* (FANG) females, processed

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in the same manner as the test samples. DNA was extracted from ELISA-positive homogenates using phenol/ethanol.[16] PCR on purified samples was done according to Snounou et al.[17] to confirm ELISA positivity.

Samples with optical densities greater than twice the average of the optical density of the negative controls were considered positive for *Plasmodium* infection. The prevalence for each *Plasmodium* species was computed as a percentage.

Identification of blood meal source using ELISA

Field-collected females were categorized as fed following examination with a hand lens. All blood-fed females had their blood meal source determined by ELISA.[18] This assay uses anti-host immunoglobulin G (IgG) conjugates directed against human, pig, dog, chicken, bovine and goat blood. Negative controls consisted of four unfed female *An. funestus* mosquitoes (FANG) and were processed as for the test samples. Positive controls included 1 μl of host blood diluted in 50 μl phosphate buffered saline and were freshly prepared as needed. Blood for the positive controls was collected by trained staff at the Department of Agriculture, University of Pretoria, Pretoria, South Africa. Test plates were read at 405 nm using an ELISA plate reader (Scientific Multiskan Ascent type 354, Thermo, Vantaa, Finland) after 30 min of substrate incubation. Samples were considered positive if their absorbance values exceeded the mean plus three times the standard deviation of the four negative controls.[18] Because of cross reactivity between immunoglobulin, data were analyzed as percentage of mosquitoes with human and animal (pig, dog, chicken, bovine and goat) blood. The blood source with the greatest percentage was considered the main blood source.

Development of hydrolysis probe assay for *Anopheles funestus* subgroup identification

Amplification and cloning of intergenic spacer region

For the development of the intergenic spacer (IGS)-based assay for the *An. funestus* group, we obtained from the National Center for Biotechnology Information (NCBI) database known 18S and 28S subunits of rDNA gene sequences for insects closely related to the *An. funestus* group: *Drosophila melanogaster* (GenBase accession no. **GI158246**), *An. gambiae* (GenBase accession no. **GI103421630**), *Aedes aegypti* (GenBase accession no. **GI169930869**) and *Culex quinquefasciatus* (GenBase accession no. **GI66269563**). Each sub-unit was aligned using Lasergene 7 MegAlign software (DNASTAR, Madison, WI, USA). Primers were preferably designed from regions with greatest sequence identity between two or more species with the aid of the Fermentas REviewer tool (inqaba biotec, Pretoria, South Africa) and synthesized (inqaba biotec, Pretoria, South Africa). Forward and reverse primers (Table 1) were respectively designed from the 28S and 18S subunits of the rDNA gene.

PCR reaction (50 μl) for IGS amplification consisted of the following: 1X LA buffer (catalogue no. KA3701DA, Takara Biomedical Group, Shiga, Japan), 0.4 mM deoxynucleotides (dNTPs), 0.5 μM of each primer pair, 2.5U of LA *Taq* DNA polymerase and 50–100 ng genomic DNA. The $MgCl_2$ concentration for reactions in which DNA from *An. funestus*, *An. rivulorum* and *An. funestus*-like was used as the template was 2 mM. That for *An. parensis* and *An. vaneedeni* was 3 mM and 2.5 mM respectively. The PCR cycling conditions were as follows: one cycle initial denaturation at 94°C for 1 min, 30 cycles of denaturation at 98°C for 5 s; annealing temperatures of 56°C for 30 s for *An. funestus* and *An. funestus*-like, 38°C for 30 s for *An. vaneedeni* and *An. rivulorum*, 46°C for 30 s for *An. parensis*; extension at 68°C for 15 s. This was followed by a final extension at 72°C for 10 min. A no-template control consisting of only PCR mix was included for each reaction to detect contamination. PCR products (10 μl) were electrophorized on a 1% Tris-acetate-EDTA (TAE) agarose gel stained with 0.5 μg/ml ethidium bromide and visualized under a UV illumination system (Syngene G-BOX, Vacutec, Johannesburg, South Africa). IGS PCR products were gel purified using QIAquick Gel Extraction Kit 28 707 (QIAGEN, Cape Town, South Africa) and cloned into pGEM-T Easy Vector A1360 (Promega, Madison, WI, USA) following the manufacturer's instructions. White clones were screened for insert by colony PCR using T7 and SP6 primers. The cycling conditions consisted of an initial denaturation at 94°C for 2 min followed by 25 cycles of denaturation at 94°C for 30 s, annealing at 56.5°C for 30 s and 72°C extension for 30 s. A final extension was performed at 72°C for 10 min. PCR products (3 μl) were

Table 1. Primers used in the amplification of the partial intergenic spacer (IGS) region of the *Anopheles funestus* group

Species	Name of primer	5′-3′primer sequence	Temp. (°C)
An. funestus	IGS Forward 1	TTC GAT CAC CTG ATG CCG C	62.00
	IGS Rev	TAC TGA GCC TTA TGC GGT TTC AC	62.77
An. vaneedeni	IGS Forward 4	CAC CTG ATG CCG CAG GTT G	64.48
An. parensis	IGS Rev 2	CTA CTT AGA CAT GCA TGG C	58.01
An. rivulorum	UN[a]	GTG TGC CCC TTC CTC GAT GT	58.3
	IGS Rev	TAC TGA GCC TTA TGC GGT TTC AC	62.77
An. funestus-like	UN18S IGS Rev	TTACTGGTAGGATCAACCAG	56
	IGS Forward 4	CAC CTG ATG CCG CAG GTT G	64.48

[a]Universal primer (UN), originally designed by Scott et al.[35]
Temp.: temperature.

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analyzed as previously described and 10 μl of the remaining product was sent for sequencing by Macrogene, Seoul, South Korea.

Sequencing analysis IGS-region

Sequencing of selected clones in both directions using T7 and SP6 primers was done on the members of the *An. funestus* subgroup. At least two different PCR products of the same size originating from the same cloning reactions were sequenced to confirm identity of the insert. Sequences were examined with the Basic Local Alignment Search Tool (BLAST searched) and those that aligned with the 18S or 28S subunits of the rDNA genes of a closely related insect to *Anopheles* were used to design the new hydrolysis probe assay.

Design of IGS-based hydrolysis probe assay for Anopheles funestus subgroup identification

Partial IGS nucleotide sequences for four species in the *An. funestus* group were aligned as previously described: *An. funestus* (GenBank accession no: **KC771136**), *An. funestus*-like (GenBank accession no. **KC771137**), *An. parensis* (GenBank accession no. **KC771138**) and *An. vaneedeni* (GenBank accession no. **KC771139**). Alignment revealed species-specific sequences from which discriminating probes were designed. Species-specific probes were designed between the universal forward and reverse primer. Probes were designed and where three or less species-specific base pair differences were present, these were designed as minor groove binder (MGB). Those probes spanning a region with more than three species-specific nucleotide differences were designed as dual-labeled probes. The 5′ end of each probe was labeled with ROX (FUN-vezen) for *An. funestus*, 6-FAM (FUN-LIKE vezen) for *An. funestus*-like, VIC (VAN-vezen) for *An. vaneedeni* and Quasar 705 (PAR-vezen) for *An. parensis* (Table 2). Universal forward and reverse primers (Table 2) specific for the targeted species were designed to produce a 403 bp product.

Primer validation and hydrolysis probe assay optimization for discrimination of An. funestus subgroup

To validate the specificity of designed primers by melt curve analysis, a PCR reaction (20 μl) consisting of 10 μl of iQ SYBR Green Supermix 170-8880 (Bio-Rad, Johannesburg, South Africa), 0.4 μl SYBR Green 1, 0.5 μM forward and reverse primers and 1 μl genomic DNA was set up. Cycling conditions comprised: 95°C for 10 min, followed by 35 cycles 95°C for 15 s, annealing at 64°C for 60 s. Included was a melt step of 72–95°C rising by 1°C and holding for 90 s for pre-melting and subsequently 5 s for each step afterwards. Changes in fluorescence of SYBR Green 1 during PCR and melting of DNA were monitored.

For the hydrolysis probes assay, PCR reactions (20 μl) contained 10 μl iQ Supermix 170-8862 (BIO-RAD, Johannesburg, South Africa), 200 nM each of FUN, FUN-LIKE and PAR probes, 25 nM of VAN probe, 0.5 μM of each primer and 1 μl of DNA template. The reactions were as follow: initial denaturation at 95°C for 10 min, followed by 35 cycles of denaturation at 95°C for 15 s, annealing at 64°C for 60 s. Increase in fluorescence of each of the reporter dyes ROX, Quasar 705, VIC and FAM was monitored in real time by acquiring each cycle on the following channels: orange (560 nm excitation and 650 nm emission), purple (684 nm excitation and 730 nm emission), green (534 nm excitation and 580 nm emission) and blue (490 nm excitation and 530 nm emission) respectively using the CFX96 Real-Time PCR Detection System (Bio-Rad, Johannesburg, South Africa).

Analytical sensitivity of assays

The allele-specific (AS)-PCR assay was used to evaluate the relative analytical sensitivity of the hydrolysis probe assay.[12] A 20 ng/ul DNA for each species was prepared and serially diluted to 1 in 10^6 dilution using nuclease-free water. The DNA preparations were used as template in the assays.

Blind trials

A blind trial was performed to validate and determine the specificity of the hydrolysis assay compared to AS-PCR. A total of 126 samples (consisting of DNA extracted from 70 *An. funestus*, 19 *An. parensis*, 14 *An. funestus*-like, four *An. rivulorum*, 12 *An. vaneedeni*, one *An. gambiae*, two *An. leesoni*, one *P. vivax* and one *P. falciparum*) and water were labeled with a code unknown to the person performing the test. Samples were identified and the results compared for both assays. *Plasmodium* species DNAs were included in the test to confirm assay specificity.

Table 2. Probes and primers used in the hydrolysis probe assay for the identification of *Anopheles funestus* subgroup

Name	Primer/probe	Reporter dye	Sequence (5'–3')	Quencher dye
FUN-vezen	Probe	ROX	TGACCGTTACACCCCAACAAATT	BHQ2[a]
FUN-LIKE-vezen	Probe	6-FAM	ACCGTTGCACCCCAACACA	MGBNFQ[b]
VAN-vezen	Probe	VIC	TCCCAATAAGCGAGTGAAATGTGT	MGBNFQ[b]
PAR-vezen	Probe	Quasar705	CCCAACAATCGTGCGAAAGTGTGT	BHQ2[a]
UN18S IGS Rev	Primer	[c]	TTACTGGTAGGATCAACCAG	[c]
IGS-Fwd UN 4	Primer	[c]	GCGAACTACCCAATGGAAC	[c]

[a] No minor groove binding.
[b] With minor groove binding.
[c] No reporter or quencher dye attached to specific primer.

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Results

Mosquito collections and species identification

In total, 250 adult *An. funestus* mosquitoes were collected: 93.6% (234/250) indoors and 6.4% (16/250) outdoors. The multiplex PCR assays identified 84.8% (212/250) of the samples (Tables 3 and 4).[4,12] The most prevalent species collected during the wet season was *An. funestus* (56.9%, 62/109), while *An. rivulorum* (73.8%, 76/103) was most prevalent during the dry season. For *An. funestus*-like, wet and dry season collections contributed 11.9% (13/109) and 22.3% (23/103) respectively (n=36). For this species, indoor collections (91.7%, 33/36) outnumbered outdoor collections (8.3%, 3/36) regardless of season.

Insecticide susceptibility assay

A total of 290 F_1 progeny from nine wild *An. funestus* s.s. females originating from Wovwe were used for susceptibility tests. Insecticide resistance was detected to bendiocarb (n=106; 62.2%, 66/106 mortality) and deltamethrin (n=120; 72.5%, 87/120 mortality). DDT exposure resulted in a 98.1% mortality (n=64, one survivor). Additional material was not available to confirm the results. No F_1 progeny of *An. funestus*-like were obtained for inclusion in the susceptibility tests.

Mosquito blood meal source identification

A total of 21 *An. funestus*-like female mosquitoes were analyzed for blood meal source and 47.6% (10/21) tested positive for animal blood (bovine and goat). The blood meal sources for 52.4% (11/21) *An. funestus*-like species could not be determined and human blood as a source was excluded. Blood meal source was only determined for *An. funestus*-like.

Plasmodium sporozoite infectivity rate

A total of 152 females were analyzed for *P. vivax* and *P. falciparum* circumsporozoite protein by ELISA. Of the samples screened, only *An. funestus*-like specimens collected during the dry season were positive for *P. vivax* (6.7%, 1/15). Pooling data from both seasons, an overall infection rate of 4% (1/25) with *P. vivax* was obtained for this species. *Anopheles funestus*-like collected in 2008 by Spillings et al.[4] from Karonga that were negative for *P. falciparum* were found positive (3.1%, n=2/64) for *P. vivax* during our study. No mixed infection of mosquitoes by different species of *Plasmodium* sporozoite was observed. Confirmation of ELISA positive samples by PCR was unsuccessful. All the *An. funestus* samples tested were negative.

Table 3. *Anopheles funestus* group species identified from wet season indoor and outdoor[a] collections

Collection site	*An. funestus*-like n (%)	*An. rivulorum* n (%)	*An. funestus* n (%)	Total (n=109)
Mwakabighili	3 (20)	12 (80)	0	15
Mwenetete	8 (42.1)	10 (52.6)	1 (5.3)	19
Wovwe	0	3 (4.8)	60 (95.2)	63
[a]Kwambwe	2 (16.7)	9 (75)	1 (8.3)	12
Total	13	34	62	109

[a] Cattle kraal collections.

RFLP genotyping

The D3 RFLP-PCR has been used in the past to genotype *An. funestus* (Garros et al., 2004; Koekemoer et al., 2006).[13,14] The same assay was used to determine if multiple genotypes were present in the *An. funestus*-like population from Karonga. Two digest patterns (genotypes) were obtained for the 23 *An. funestus*-like samples processed. To differentiate these new genotypes from those of *An. funestus* they were designated as Type 1 and Type 2 (Figure 2).

Type 1 genotype constituted 95.7% (22/23) of collected *An. funestus*-like samples, while Type 2 was in the minority (4.3%, 1/23). Sequencing confirmed electrophoresis patterns and Type 1 genotype amplifies one fragment (375 bp) that is digested into three fragments (70 bp, 148 bp and 157 bp long). Type 2 genotype amplifies two products (399 bp and 337 bp) that are then digested. The 339 bp fragment is digested into three fragments (83 bp, 150 bp and 166 bp) while the 337 bp product is digested into 150 bp, 75 bp and 112 bp fragments (Figure 3).

Of the 60 *An. funestus* samples analyzed, 51.6% (31/60) of samples were Type Y and 43.3% (26/60) were Type MW; only 5% (3/60) of samples were Type M (Figure 3) as described by Koekemoer et al.[14] One *An. funestus* sample showed a new, unknown genotype (Figure 2) and is provisionally named Type X.

IGS region amplification and design of hydrolysis probe assay

Amplified IGS-PCR products were cloned and sequenced to confirm insert identity. The 3′-region of the IGS sequence was successfully obtained for *An. funestus*, *An. funestus*-like, *An. vaneedeni* and *An. parensis*. Alignment of the 3′ end sequence of the IGS region for the four species in the *An. funestus* group is shown in Figure 4. The limited species-specific sequences available in this region meant that it was not possible to develop a multiplex conventional PCR. Primers and probes were designed and a hydrolysis probe assay (TaqMan gene expression assay) developed. PCR conditions were optimized using known positive control DNA templates extracted from *An. funestus*, *An. funestus*-like, *An. vaneedeni* and *An. parensis*. An example of the identification of seven to ten samples per species is showed in Figure 5. Increase in fluorescence of the ROX, Quasar 705, VIC and 6FAM probes indentifies *An. funestus*, *An. parensis*, *An. vaneedeni* and *An. funestus*-like respectively.

Analytical sensitivity

The analytical sensitivity of the hydrolysis probe assay was determined using serial dilutions of DNA extracted from representative species. The DNA concentration with the lowest dilutions to produce a quantitative cycle (Cq) value above the threshold of the negative control in the hydrolysis probe assay was compared with the lowest dilutions to produce an observable amplification

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Table 4. *Anopheles funestus* group species identified from dry season indoor and outdoor[a] collections

Collection site	*An. funestus*-like n (%)	*An. rivulorum* n (%)	*An. funestus* n (%)	*An. parensis* n (%)	Total (n=103)
Mwakabighili	3 (15)	17 (85)	0	0	20
Mwenetete	13 (19.1)	52 (76.5)	1 (1.5)	2 (2.9)	68
Wovwe	0	0	1 (100)		1
Kwambwe[a]	1 (100)	0			1
Mwampaghatwa	6 (46.2)	7 (53.8)			13
Total	23	76	2	2	103

[a]Cattle kraal collections.

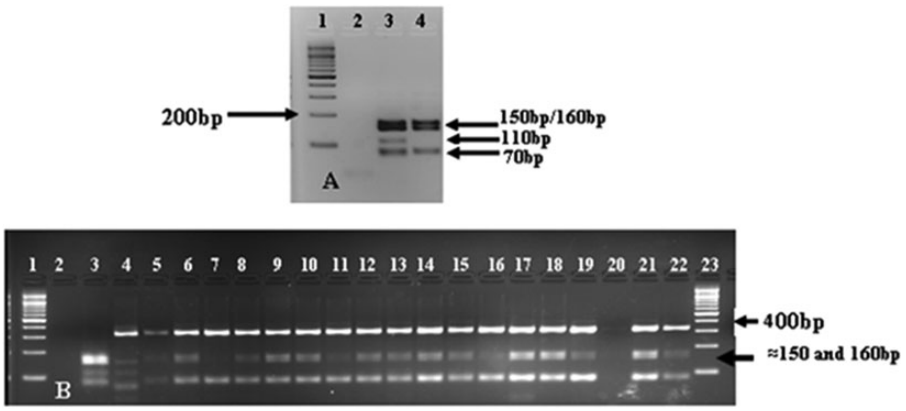


Figure 2. RFLP genotype analysis on *An. funestus*-like and *An. funestus*. (A) *Anopheles funestus*-like samples after *Hpa* II digestion. Lane 1: DNA molecular marker, Lane 2: no-template negative control, Lane 3: Type 2, Lane 4: Type 1. (B) *Anopheles funestus* s.s samples post *Hpa* II digestion. Lane 1: DNA molecular marker, Lane 2: no-template negative control, Lane 3: Type X; Lanes 7 and 11: Type M; Lanes 5, 6, 8, 9, 10, 12–16, 19 and 22: Type Y; Lanes 17, 18 and 21, Type MW. Lane 20: no amplification.

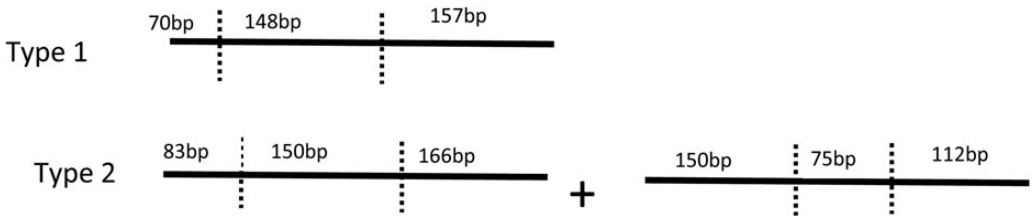


Figure 3. Schematic representation of rDNA D3 *Hpa* II restriction mapping for different restriction fragment length polymorphism types. The black bold lines represent DNA (D3 PCR products) and the dotted vertical lines indicate the *Hpa* II enzyme restriction sites. Each restriction site is 4bp and, together with the primer length, was not considered when constructing the restriction map.

product on agarose gel for AS-PCR. Results indicated that the detection limit of the hydrolysis probe assay for all species was 1 in 1000 (1/1000) dilution (0.02 ng of DNA) while the AS-PCR was only able to amplify with the 1/500 dilutions (0.04 ng of DNA) (results not shown).

Blind trial of assay

The performance of the AS-PCR and hydrolysis probe assay using DNA extracted from mosquitoes collected across different regions in sub-Saharan Africa is summarized in Table 5. Results indicate that the hydrolysis probe assay had a lower failure rate of 12.7% (16/126) compared to 21.4% for AS-PCR (27/126). The DNA samples from *Plasmodium* species, *An. gambiae*, *An. leesoni* and *An. rivulorum* together with water gave negative results for both assays.

Discussion

The biology and resting behavior of *Anopheles funestus*-like was investigated and collections in Karonga revealed a relatively low

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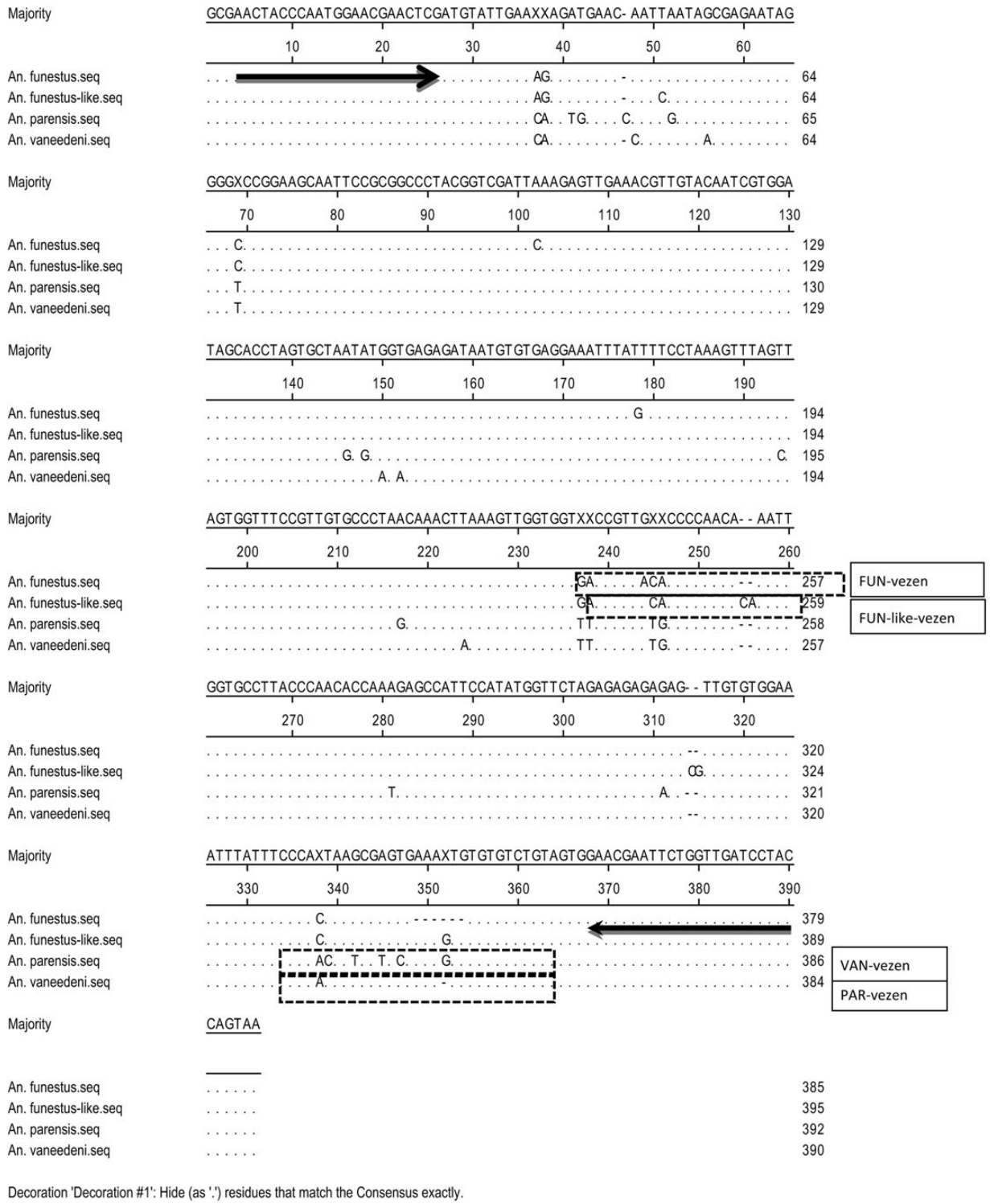


Figure 4. Alignment of partial intergenic spacer (IGS) sequences for members of *the Anopheles funestus* subgroup including *An. funestus*-like, showing position of species-specific probes and primers used for designing a hydrolysis probe assay. Conserved nucleotides are shown in dots, deletions in dashes. Solid black arrows indicate the position of the IGS-Fwd UN4 and UN 18S IGS Rev primers. Positions of probes are indicated with boxes at relevant positions and names of probe are indicated on the right hand side of the graph.

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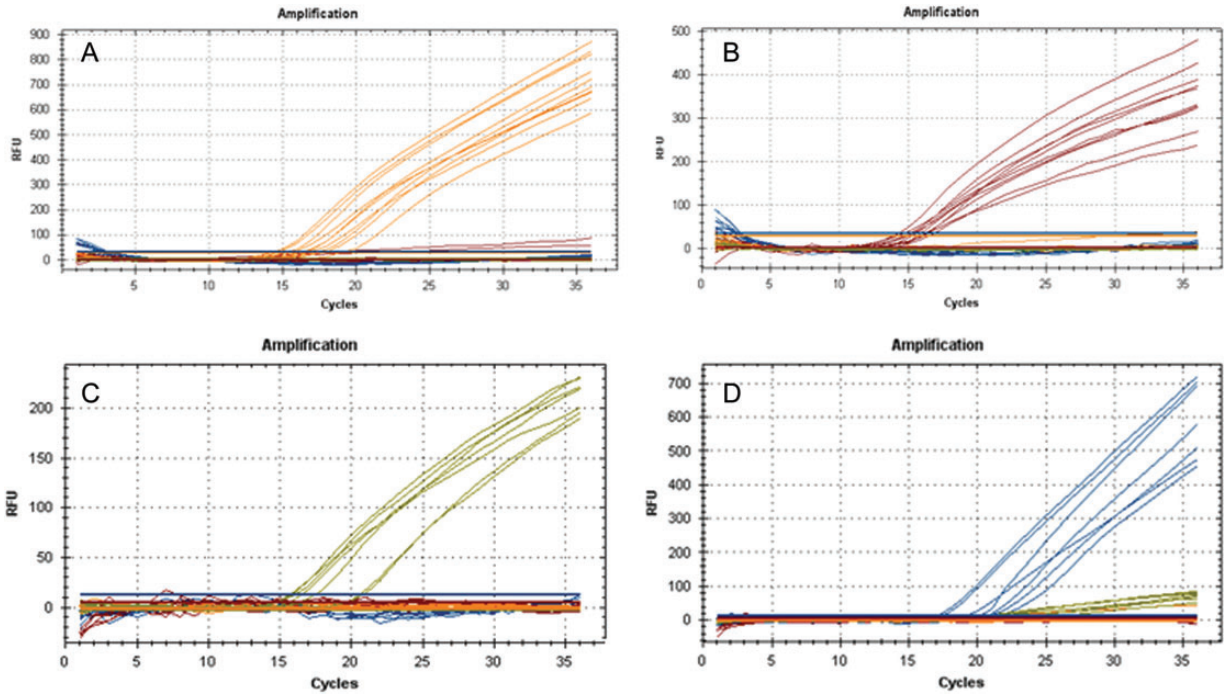


Figure 5. Identification of members of the *An. funestus* subgroup using a hydrolysis probe assay based on the intergenic spacer (IGS) of rDNA gene. Seven to ten specimens of *An. funestus* (A), *An. parensis* (B), *An. vaneedeni* (C) and *An. funestus*-like (D) were tested. (A) Cycling of the FUNvezen probe (ROX labeled), (B) cycling of the PARvezen probe (Quasar 705 labeled), (C) cycling of the VANvezen probe (VIC labeled) and (D) cycling of the FUN-LIKEvezen probe (6FAM labeled).

Table 5. Performance of the allele-specific (AS)-PCR and hydrolysis probe assays in a blind trial for the identification of *Anopheles funestus* subgroup species

	AS-PCR (n=26)	Hydrolysis probe assay (n=126)
Correct identification	99	110
Failed amplification	27	16
Mis-identification	0	0

number of mosquitoes. The low sample size places limitations on interpretation of the data, but interesting findings were observed. *Anopheles funestus*-like was predominantly collected resting indoors, in line with previous reports by Spillings et al.[18] Although *An. funestus*-like species were collected mostly inside occupied houses, blood meal analysis showed that this species is mainly feeding on animals. This observation suggests that the mosquitoes obtained a blood meal outdoors, from animals, before entering houses to rest. This behavior may be to evade high outdoor temperatures, and such feeding behavior has been reported among other members of the *An. funestus* group.[19,20,21] Failure to collect *An. funetsus*-like by human landing catch, together with the blood meal analysis data, suggests that this species is zoophilic. The number of unidentified blood meals was high and unexpected, indicating that other hosts such as birds and rats might be involved. This highlights the need to include a wider array of hosts for blood meal source determination and more sensitive assays such as PCR or sequencing to accurately identify blood meal source.[22]

The feeding behavior of mosquitoes is associated with their potential to transmit malaria parasites. Spillings et al. provide the most recent *Plasmodium* infectivity study in Karonga from wet season collections.[4] In their study, field collected adult mosquitoes (*An. rivulorum* and *An. funestus*-like) were screened for *P. falciparum* infection and all specimens tested negative. *Anopheles funestus*-like specimens tested in our study were screened for both *P. falciparum and P. vivax* and again there was an absence of *P. falciparum* infection. A surprising result in our study was that *An. funestus*-like was positive for *P. vivax* sporozoites, even though no evidence was found that it fed on humans. However, it is most likely that the CSP-ELISA produced false positive results as previously reported for other anopheline species.[16,21,23,24] Repeated attempts to confirm the sporozoite ELISA results using PCR were unsuccessful, but factors such as human error, DNA degradation or low *P. vivax* template could not be excluded. Further investigation into the infection of *An. funestus*-like by *P. vivax* using a more sensitive technique such as real-time PCR,[25] which uses probes specific to the *Plasmodium* parasite, is recommended. ELISA-positive samples can be confirmed by a second ELISA using a heated lysate of the positive samples,[24] but we had insufficient ELISA homogenate to do this in our study. No data were available to us to confirm the presence of *P. vivax* from the villages at the study site and this should be investigated in future studies. The importance of *P. vivax* in malaria transmission is often under reported.

Four members of the *An. funestus* group were identified in this study: *An. parensis, An. rivulorum, An. funestus*-like and *An.*

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funestus. Irrespective of the collection season, *An. rivulorum* was found to be the predominant species in Mwakabighili, Mwenetete and Kwambwe while *An. funestus* was the predominant species in Wovwe. The sympatric occurrence of members of the *An. funestus* group in this area has been reported before (Chiphwana, unpublished data) and highlights the importance of accurate species identification.[4]

Molecular analysis of the *An. funestus*-like and *An. funestus* samples revealed the presence of different molecular forms. The D3-RFLP PCR assay confirmed the molecular genotypes present in Malawi for *An. funestus*. This is the first report of molecular typing of *An. funestus*-like and two genotypes were observed. This supports the existence of two genetic clades as described by Choi et al.[26] The relationship between the molecular clades and the molecular RFLP forms will need to be clarified in future. The overlap in fragment profiles between *An. funestus* and *An. funestus*-like means that molecular typing should not be used as an identification tool to differentiate between the two species. However, such typing does provide additional information with regard to the population polymorphisms present in a specific area. One *An. funestus* specimen was of an unknown genotype and analysis is needed to investigate this further.

Anopheles funestus-like is morphologically similar to the *An. funestus* subgroup, which includes species such as *An. funestus*, *An. vaneedeni* and *An. parensis*. A molecular assay is available to differentiate between members of the *An. funestus* subgroup and *An. funestus*-like.[4] This assay requires two separate PCR steps before *An. funestus*-like can be identified and alternative methods were investigated. A novel assay was developed to differentiate between these species. Partial 3′-end IGS sequences of the rDNA were obtained for the four species of the *An. funestus* subgroup. Difficulties in obtaining IGS sequences from the 5′ end of the rDNA gene could have been the result of secondary structures inhibiting the sequencing reaction. The possibility of developing a cost-effective AS-PCR based on the IGS sequences was investigated but the technique was not specific enough, hence the development of a hydrolysis-probe assay. The strength of this assay is that it can be used to discriminate *An. funestus*-like from the other members in the group where morphological identification is highly similar.

The analytical sensitivity and robustness of the hydrolysis probe assay exceeds that of conventional AS-PCR. This finding is in line with previous reports, in which real-time assays were developed for the detection of mutations implicated in insecticide resistance,[27] discrimination of malaria parasites[25] and identification of malaria vectors.[28,29] This sensitivity can be attributed to the double specificity resulting from the use of species-specific probes and primers. The specificity of the assay in the blind test was comparable to that of AS-PCR, as no samples were misidentified. Further analysis of blind trial results showed that the hydrolysis probe assay identified more *An. funestus* subgroup DNA samples than AS-PCR. It was noticed that some of the samples identified using the hydrolysis probe assay and not by AS-PCR, had concentrations in excess of 100 ng/μl. This suggests a wider dynamic range for the hydrolysis probe assay over the AS-PCR.

The *An. funestus* population from Wovwe was resistant to deltamethrin and bendiocarb but susceptible to DDT. The underlying resistance mechanism in this population is unlikely to be knock-down resistance (*kdr*) mutations since *kdr* normally confers resistance to pyrethroids and DDT.[30,31] The observed resistance profile in Wovwe tallies with previous reports from southern Africa[2,3,32,33] but differs from East Africa records. In Uganda, Morgan et al.[34] reported susceptibility to bendiocarb and deltamethrin and suspected resistance to DDT in *An. funestus*. This study highlights the possible spread of resistant genes in *An. funestus* populations from southern Africa and emphasizes the need to collaborate and incorporate insecticide resistance management strategies in the various malaria control programs within the region.

The conclusions that can be drawn from this study are limited by the small sample size collected during the study period. Additional collections are needed of *An. funestus*-like to verify results obtained in this study.

Conclusion

The dynamics of malaria transmission in the Karonga region of northern Malawi are poorly understood. Understanding the vector species composition, pattern of malaria transmission, prevalence of *Plasmodium* infection and the insecticide susceptibility status of target vector populations is critical in the design of a co-ordinated malaria control program. This study highlights the presence of four species of the *An. funestus* group (*An. funestus*, *An. rivulorum*, *An. parensis* and *An. funestus*-like) in Karonga. Previous methods failed to identify *An. funestus*-like species, and our hydrolysis probe assay provides a means to discriminate members of the *An. funestus* subgroup including *An. funestus*-like in a single reaction. Although the analytical specificity of this assay is comparable to that of the gold standard PCR for *An. funestus* group identification, it is a more sensitive assay with a greater dynamic range. Furthermore, the D3-RFLP genotyping can differentiate two molecular forms in *An. funestus*-like. Although this species is shown to be zoophilic and mainly endophilic, its vector status in terms of the malaria transmission needs to be verified on a larger sample size as well as over time.

Authors' contributions: The study was conceived by RH, MC, CB and LK. SV, JC and RH performed field collection. SV performed all laboratory assays, analyzed results and wrote the initial manuscript. All authors read and approved the final manuscript. LK is guarantor of the paper.

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References

1 President's Malaria Initiative: Malaria Operational Plan, Malawi. 2011. Available at http://pmi.gov/countries/mops/fy12/malawi_mop_fy12.pdf. [accessed April 2013].

2 Hunt RH, Edwardes M, Coetzee M. Pyrethroid resistance in southern African *Anopheles funestus* extends to Likoma Island in Lake Malawi. Parasit Vectors 2010;3:122.

3 Wondji CS, Coleman M, Kleinschmidt I et al. Impact of pyrethroid resistance on operational malaria control in Malawi. Proc Natl Acad Sci U S A 2012;109:19063–70.

4 Spillings B, Brooke BD, Koekemoer LL et al. A new species concealed by *Anopheles funestus* Gilies, a major malaria vector in Africa. Am J Trop Med Hyg 2009;81:510–15.

5 Kwapata K, Mwase WF, Bokosi JM et al. Genetic diversity of *Annona senegalensis* Pers. Populations as revealed by simple sequence repeats (SSRs). Afr J Biotech 2007;6:1239–47.

6 Hunt RH, Brooke BD, Pillay C et al. Laboratory selection for and characteristics of pyrethroid resistance in the malaria vector *Anopheles funestus*. Med Vet Entomol 2005;19:271–5.

7 WHO. Test procedures for insecticide resistance monitoring in malaria vectors. 2013. Available at http://apps.who.int/iris/bitstream/10665/80139/1/9789241505154_eng.pdf. [accessed April 2013].

8 Abbott WS. A method of computing the effectiveness of an insecticide. J Econ Entomol 1925;18:265–6.

9 Gillies MT, De Meillon B. The Anophelinae of Africa South of the Sahara. Publications of the South African Institute for Medical Research, Johannesburg. 1968;54:1–341.

10 Gillies MT, Coetzee M. A supplement to the Anophelinae of Africa South of the Sahara. Publications of the South African Institute for Medical Research, Johannesburg. 1987;55:1–141.

11 Collins F, Mendez AM, Rasmussen MO et al. A ribosomal RNA gene probe differentiates member species of the *Anopheles gambiae* complex. Am J Trop Med Hyg 1987;37:37–41.

12 Koekemoer LL, Kamau L, Hunt RH, Coetzee M. A cocktail polymerase chain reaction assay to identify members of the *Anopheles funestus* (Diptera: Culicidae) group. Am J Trop Med Hyg 2002;66:804–11.

13 Garros C, Koekemoer LL, Kamau L et al. Restriction fragment length polymorphism method for the identification of major African and Asian malaria vectors within the *Anopheles funestus* and *An. minimus* groups. Am J Trop Med Hyg 2004;70:260–5.

14 Koekemoer LL, Kamau L, Garros C et al. Impact of the Rift Valley on restriction fragment length polymorphism typing of the African malaria vector *Anopheles funestus* (Diptera: Culicidae). J Med Entomol 2006;43:1178–84.

15 Wirtz RA, Sattabonkot J, Hall T et al. Development and evaluation of an ELISA for *Plasmodium vivax*-VK247 sporozoites. J Med Entomol 1992;29:854–57.

16 Koekemoer LL, Rankoe EM, La Grange JP et al. False detection of *Plasmodium falciparum* sporozoite in *Anopheles marshallii* group of mosquitoes. J Am Mosq Contr Assoc 2001;17:160–65.

17 Snounou G, Viriyakosol S, Zhu XP et al. High sensitivity of detection of human malaria parasite by the use of nested polymerase chain reaction. Mol Biochem Parasitol 1993;61:315–20.

18 Beier JC, Perkins PV, Wirtz RA et al. Blood–meal identification by direct enzyme-linked immunosorbent assay (ELISA) tested on *Anopheles* (Diptera: Culicidae) in Kenya. J Med Entomol 1988;25:9–16.

19 Kamau L, Koekemoer LL, Hunt RH, Coetzee M. *Anopheles parensis*: the main member of the *Anopheles funestus* species group found resting inside human dwellings in Mwea area of central Kenya towards the end of the rainy season. J Am Mosq Contr Assoc 2003;19:130–33.

20 Kent RJ, Norris DE. Identification of mammalian blood meals in mosquitoes by a multiplexed polymerase chain reaction targeting cytochrome b. Am J Trop Med Hyg 2005;73:336–42.

21 Mouatcho JC, Hargreaves K, Koekemoer LL et al. Indoor collections of the *Anopheles funestus* group (Diptera: Culicidae) in sprayed houses in northern KwaZulu-Natal, South Africa. Malar J 2007;6:30.

22 Kent RJ, Thuma PE, Mharakurwa S, Norris DE. Seasonality, blood feeding behaviour, and transmission of *Plasmodium falciparum* by *Anopheles arabiensis* after an extended drought in southern Zambia. J Trop Med Hyg 2007;76:267–74.

23 Durnez L, Van Bortel W, Denis L et al. False positive circumsporozoite protein ELISA: a challenge for the estimation of the entomological inoculation rate of malaria and for vector incrimination. Malar J 2011;10:195.

24 Lochouarn L, Fontenille D. ELISA detection of malaria sporozoites: false-positive results in *Anopheles gambiae* s.l. associated with bovine blood meals. Trans R Soc Trop Med Hyg 1999;93:101–2.

25 Bass C, Nikou D, Blagborough AM et al. PCR-based detection of *Plasmodium* in *Anopheles* mosquitoes: a comparison of a new high-throughput assay with existing methods. Malar J 2008;7:177.

26 Choi KS, Koekemoer LL, Coetzee M. Population genetic structure of the major malaria vector *Anopheles funestus* s.s. and allied species in southern Africa. Parasit Vectors 2012;5:283.

27 Bass C, Nikou D, Donnelly M et al. Detection of knockdown resistance (*kdr*) mutations in *Anopheles gambiae*: a comparison of two new high-throughput assays with existing methods. Malar J 2007a;6:111.

28 Bass C, Williamson MS, Wilding CS et al. Identification of the main malaria vectors in the *Anopheles gambiae* species complex using a TaqMan real-time PCR assay. Malar J 2007b;6:155.

29 Vezenegho SB, Bass C, Puinean M et al. Development of multiplex real-time PCR assays for identification of members of the *Anopheles funestus* species group. Malar J 2009;8:282.

30 Martinez-Torres D, Chandre F, Williamson MS et al. Molecular characterisation of pyrethroid knockdown resistant (*kdr*) in the major malaria vector, *Anopheles gambiae* s.s. Insect Mol Biol 1998;7:179–84.

31 Dabire KR, Diabaté A, Namountougou M et al. Distribution of pyrethroid and DDT resistance and the L1014F *kdr* mutation in *Anopheles gambiae* s.l. from Burkina Faso (West Africa). Trans R Soc Trop Med Hyg 2009;103:1113–20.

32 Abilio AP, Kleinschmidt I, Rehman AM et al. The emergence of insecticide resistance in central Mozambique and potential threat to the successful indoor residual spraying malaria control programme. Malar J 2010;10:110.

33 Kloke RG, Nhamahanga E, Hunt RH, Coetzee M. Vectorial status and insecticide resistance of *Anopheles funestus* from a sugar estate in southern Mozambique. Parasit Vectors 2011;4:16

34 Morgan JC, Irving H, Okedi LM et al. Pyrethroid resistance in an *Anopheles funestus* population from Uganda. PLoS One 2010;5:e11872.

35 Scott JA, Brogdon WG, Collins FH. Identification of single specimens of the *Anopheles gambiae* complex by the polymerase chain reaction. Am J Trop Med Hyg 1993;49:520–29.

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